

BHARATH SINGH PALTHYAVATH

bharathpalthyavath@gmail.com — +91 8688866136 — LinkedIn — GitHub — LeetCode — Portfolio

Profile

Aspiring AI/ML Engineer with hands-on experience in Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG), and Explainable AI. Built end-to-end AI applications for fraud detection and recruitment intelligence using FastAPI, LangChain, vector databases, and machine learning models. Passionate about developing scalable AI solutions and applying data-driven techniques to solve real-world problems.

Education

Chaitanya Bharathi Institute of Technology, Hyderabad	2023 – 2027
B.E in Artificial Intelligence and Data Science	CGPA: 8.92/10
Prathibha Junior College, Mahabubnagar	2021 – 2023
MPC	Marks: 962/1000
Krishnaveni Talent School, Pargi	2020 – 2021
SSC	GPA: 10/10

Experience

Machine Learning Intern — Cognifyz Technologies Jun 2025 – Jul 2025

- Built end-to-end ML pipelines including preprocessing, feature engineering, model training and evaluation.
- Conducted exploratory data analysis and created insights using Python, Pandas and Matplotlib.
- Containerized models using Docker improving deployment reproducibility.

Projects

GraphShield AI — Graph Neural Network-Based Fraud Detection System Jun 2026
GitHub Link

- Developed a fraud detection platform using Graph Neural Networks (GraphSAGE) to identify suspicious banking transactions and hidden fraud patterns.
- Built a FastAPI backend and Streamlit dashboard for real-time fraud prediction, investigation, and explainable AI using SHAP.
- Created synthetic banking datasets and achieved 99.55% recall with 0.9985 ROC-AUC on the evaluation dataset.

RecruitIQ — AI-Powered Recruitment Intelligence Platform Jun 2026
GitHub Link

- Engineered an intelligent hiring platform that automates resume parsing, candidate classification, ranking, and job-description matching using Machine Learning, NLP, and predictive analytics.
- Implemented RAG-based interview intelligence using embeddings, vector databases, semantic retrieval, and LLMs to generate role-specific interview questions and evaluate candidate responses.
- Developed explainable candidate recommendations with SHAP, multi-resume processing, recruiter analytics, and automated reporting through a scalable FastAPI-based architecture.

AI Research Paper Assistant & Academic Mentor Jun 2026
GitHub Link

- Developed a locally hosted AI-powered research assistant using LLM, FAISS, LangChain, and Streamlit for intelligent academic paper analysis and conversational Q&A.
- Built an end-to-end RAG pipeline that extracts PDF content, generates embeddings, creates FAISS vector indexes, and enables source-cited question answering across single and multiple research papers.
- Implemented automated PowerPoint presentation generation with academic templates, speaker notes, figure extraction, flowchart creation, literature review synthesis, research gap identification, and AI-driven project roadmap generation.

Technical Skills

Languages: Python, Java, SQL, JavaScript, R

Data Science & AI: Machine Learning, Regression, Classification, Clustering, Feature Engineering, Data Preprocessing, Model Evaluation, EDA, Predictive Modeling, SHAP

Generative AI: Prompt Engineering, RAG, LLMs, LangChain, Embeddings, Semantic Search, Vector Databases (FAISS, ChromaDB), AI Agents

Data Analytics: Pandas, NumPy, Excel, Power BI, Data Visualization

Backend Development: FastAPI, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, VS Code, Jupyter Notebook

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

Achievements

Runner-Up — Cyber Fest Hackathon, CBIT	Feb 2026
--	----------

Certifications

Python for Data Science — NPTEL	Oct 2025
MongoDB Python Developer — MongoDB University	Apr 2025
Cyber Security & Forensics Workshop — CSI	Jan 2026
Microsoft Azure Fundamentals (AZ-900) — Microsoft	Apr 2026

Position of Responsibility

Volunteer — Chaitanya Spandana	Jan 2024 – Present
Volunteer — InsightX	Feb 2024, Feb 2025